Prasang Das

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.41
Holy Cross School Agartala	Class XII (ISC)	2024	91.50%
Holy Cross School Agartala	Class X (ICSE)	2022	92.70%

RESEARCH EXPERIENCE

Undergraduate Research Assistant | Deep RL for Graph-Structured Sequential Optimization ***(****Advisor: Dr. Sangkha Borah, IIT Hyderabad****)***

- Designed a **GNN policy** using edge-**conditioned Graph Attention (GAT)** and **message-passing (MPNN)** layers that fold structural and continuous edge features into attention, enabling the actor-critic to reason natively over **pairwise O(N^2) action spaces.**
- Developed a curriculum-learning schedule with rate-limiting and two-sided (promote/demote) transitions to stabilize training as task difficulty scales, plus reward shaping to guide the agent toward long-horizon strategies under sparse terminal rewards.

PROJECTS

Quantum Phase Recognition Pipeline [Github]

- Engineered a hybrid Qiskit + JAX architecture: Qiskit for Hamiltonian/circuit specification (SparsePauliOp, parameterized circuits) and a custom differentiable JAX statevector simulator for execution, enabling exact jax.grad gradients instead of black-box primitives; verified against Qiskit's Statevector to machine precision.
- Modeled realistic hardware noise (T1/T2 thermal relaxation + readout error via qiskit-aer) and implemented Zero-Noise Extrapolation recovering ground-state energies with a 3.6× reduction in error vs. the raw noisy estimate.

Time Independent Schrodinger Equation PINN [Github]

- Engineered a **PyTorch PINN eigensolver** that recovers quantized energies and wavefunctions for **three canonical Hamiltonians** (harmonic oscillator, infinite well, anharmonic Morse), enforcing boundary conditions as hard constraints and computing energies through a **differentiable Rayleigh-quotient loss.**
- Devised a **sequential orthogonality scheme** to isolate excited states across **principal quantum numbers**, refining with an **Adam then L-BFGS** schedule in float64 to match analytic spectra within **~1e-6 at ground and under 1% error overall**, with a reproducible notebook that visualizes wavefunctions and densities.

Hierarchical Graph-RAG System for Scientific Literature [Github]

- Engineered a **full-stack Retrieval-Augmented Generation (RAG)** web application **(FastAPI, Next.js)** that asynchronously **parses raw LaTeX from arXiv** to construct and visualize massive **academic citation networks.**
- Trained **unsupervised Graph Neural Networks (GAT via JAX)** and combined them with instruction-tuned semantic embeddings to drive **a local 31B params LLM (Gemma4)** that synthesizes highly-accurate literature reviews.

Pneumonia-Detection-CNN Python, PyTorch, Pandas, Deep Learning [Github]

- Developed a PyTorch-based **Convolutional Neural Network (CNN) for chest X-ray pneumonia detection**, implementing custom pipelines, workflows, checkpointing, and evaluation, achieving **85% F1-score and 98% recall**.
- Implemented **Grad-CAM explainability** and error analysis to visualize model decision-making, analyze false positives/false negatives, and assess model interpretability in medical image classification tasks.

Spectral Error Mitigation for Distributed Field Simulations [Github]

- Engineered a 1D Klein-Gordon field simulation implementing redundancy-based Quantum Error Correction (QEC) principles to achieve a **5x improvement** in numerical fidelity while **reducing L^2 error and cumulative energy drift** by nearly an **order of magnitude across long-time evolution.**
- Architected a **hybrid QFT-based diagnostic framework** to isolate high-frequency stochastic noise in relativistic simulations, successfully stabilizing numerical evolution and suppressing relative energy drift from **>10^5% to**

<10^4%

SKILLS

Programming: Python, C/C++, Java, LaTeX, bash

Libraries & Frameworks: PyTorch, JAX, Scikit-learn, NumPy, SciPy, Matplotlib, Qiskit, QuTiP, FAISS

Tools & OS: Git, Linux (Arch and Ubuntu), Windows, Fusion 360

RELEVANT COURSES

Physics & Engineering: Computational Physics, Quantum Mechanics, Analog Electronics, Classical Mechanics

Computer Science & Mathematics: Introduction to Programming, Data Structures and Algorithms, Introduction to AI, Linear Algebra, Introduction to Probability, Introduction to Statistics, Basic Calculus

POSITIONS OF RESPONSIBILITY

Head, Behind the Lens - The Filmmaking Club of IIT Hyderabad

Head, PRO and Ideation, Institute Technical Council, IIT Hyderabad

- Serving as the Head of **Public Relations** to impact campus reach and exposure of Tech on/off campus
- Directed technical programs, conducted events, and managed public relations for the **Thrust IIT Hyderabad**

Coordinator, Outreach and Collaborations, TEDxIITHyderabad

- Conducted and managed public relations and collaborative outreach for TEDxIITHyderabad 10th edition.

EXTRACURRICULAR

Inter IIT Cultural Meet 8.0 Participant, as film editor for IIT Hyderabad team

Lead Guitarist, Performed at college events and coordinated live band sessions.